YOLO v10

Input for YOLOv10 training:

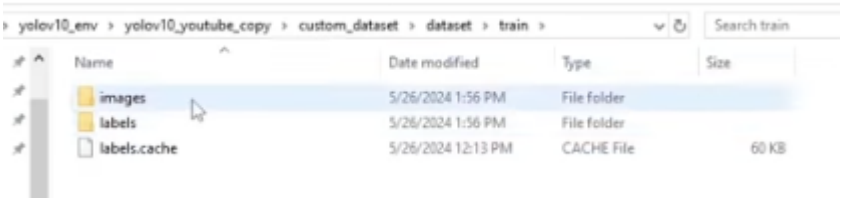
1. Images: A dataset of labeled images.
2. Annotations: Bounding box coordinates and class labels for objects in the images.
3. Configuration file: Model architecture, hyperparameters, path, and training settings.

Output during YOLOv10 training:

```
Validating runs/detect/train/weights/best.pt...
Ultralytics YOLOv8.2.49 🚀 Python-3.10.12 torch-2.3.0+cu121 CUDA:0 (Tesla T4, 15102MiB)
YOLOv10n summary (fused): 285 layers, 2694806 parameters, 0 gradients, 8.2 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%|          |
                   all         13         56       0.96      0.964      0.983      0.823
Speed: 0.2ms preprocess, 4.2ms inference, 0.0ms loss, 0.1ms postprocess per image
Results saved to runs/detect/train
ultralytics.utils.metrics.DetMetrics object with attributes:
```

P - Precision
R - Recall

1. Loss values: Overall loss and component losses (e.g., classification, objectness (confidence), bounding box regression).
2. Metrics: mAP (mean Average Precision), precision, recall, etc.
3. Trained model weights: Updated at regular intervals and at the end of training.
4. Training progress: Typically includes epoch number, learning rate, and time elapsed.



Similarly, validation and test folders follows the same directory organization (pictures with the corresponding la.

Input Files (Images/Videos)

- **Images:**
  - Common image formats used as input are `.jpg`, `.jpeg`, `.png`, and `.bmp`.
  - These images are used for training, validation, and inference.
- **Videos:**
  - Supported video formats include `.mp4`, `.avi`, `.mov`, and `.mkv`.

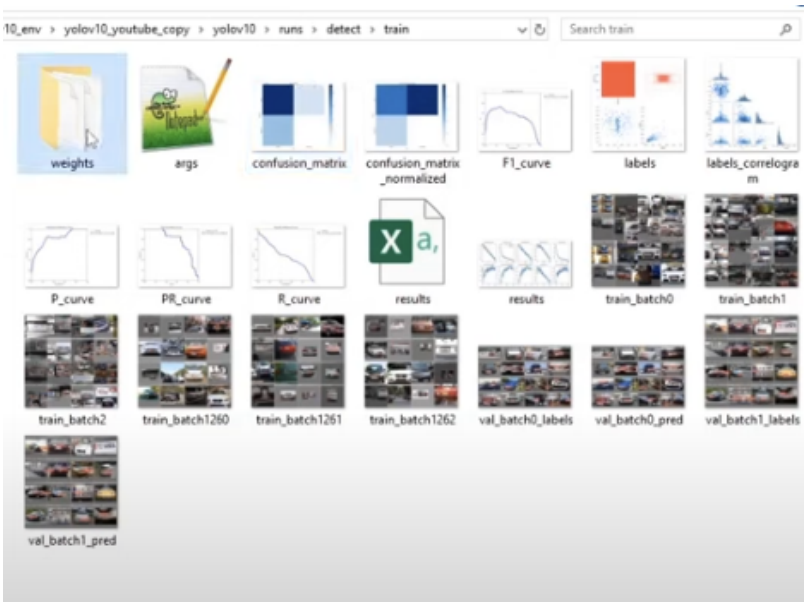
- Video inputs are often used for object detection in a sequence of frames.

Training:

```
!yolo task=detect mode=train epochs=100 batch=16 plots=True model=weights/yolov10n.pt data=custom_data.yaml
```

Inside yolov10>runs>detect>train>weights The best and last weights can be found

In the train folder itself, we can find the results, confusion matrix etc.



- Annotations in YOLOv10 are usually stored in **text files** with the `.txt` extension. Each image has a corresponding `.txt` file that holds the annotation data.
- The annotations for an image with the filename `image1.jpg` would be stored in `image1.txt`.
- These images are saved by default in a folder under `runs/predict/`, along with the class labels and confidence scores
- YOLOv8 typically uses a `classes.txt` file that contains the names of the object classes used for annotation.
- This file simply lists the class names line by line, where the line number corresponds to the `class_id` in the annotation files..
  Annotations:
  YOLO annotations are typically stored in plain text files with a `.txt` extension, where each file corresponds to one image in your dataset. The format is:

```
<class_id> <x_center> <y_center> <width> <height>
```

Trained Model Weights

- **Description**: The `.pt` file generated after training, typically named `best.pt` or `last.pt`.

- **Usage**: Contains the trained parameters of your model for inference.

Source Data

- **Description**: The images or videos you want to perform object detection on.
- **Options**:
  - **Single Image**: `source=/path/to/image.jpg`
  - **Directory of Images**: `source=/path/to/images/`
  - **Video File**: `source=/path/to/video.mp4`
  - **Webcam**: `source=0` (default webcam)

Explanation of Values:

- **class_id**: The class label index (starting from 0 for the first class).
- **x_center**: The normalized x-coordinate of the center of the bounding box (value between 0 and 1, relative to image width).
- **y_center**: The normalized y-coordinate of the center of the bounding box (value between 0 and 1, relative to image height).
- **width**: The normalized width of the bounding box (value between 0 and 1, relative to image width).
- **height**: The normalized height of the bounding box (value between 0 and 1, relative to image height).
- When a video is processed, YOLOv10 outputs a new video with bounding boxes, labels, and confidence scores displayed on each frame.
- **Output Format**: The output video is saved in `.mp4` format by default.

6. Configuration Files (YAML)

- YOLOv10 uses a `.yaml` file for dataset configuration.
- Example of a `data.yaml` file: `train: ./datasets/images/train val: ./datasets/images/val nc: 80 names: ['person', 'bicycle', 'car', 'motorbike', 'aeroplane', ...]` This file defines:
  - `train`: Path to the training image set.
  - `val`: Path to the validation image set.
  - `nc`: Number of classes.
  - `names`: List of class names.
- Given below is the data organization

```
dataset/
├── images/
│   ├── train/
│   │   ├── image1.jpg
│   │   ├── image2.jpg
│   │   └── ...
│   └── val/
│       ├── image1.jpg
│       ├── image2.jpg
│       └── ...
├── labels/
│   ├── train/
│   │   ├── image1.txt
│   │   ├── image2.txt
│   │   └── ...
│   └── val/
│       ├── image1.txt
│       ├── image2.txt
│       └── ...
├── data.yaml
```